

3 Methodology

3.1 Study design

This study employs a retrospective observational design integrating child-level data from the Multiple Indicator Cluster Survey (MICS) 2019 (or 2025, if available) with district-level agricultural production statistics and climate data. Climate and agricultural exposures will be reconstructed for each child during the first 1,000 days of life, defined as the period from conception to 24 months after birth (Victora et al., 2008). Month and year of birth will be derived from child age in months and survey timing to construct precise exposure windows. Agricultural production and climate shocks occurring within this biologically critical window will be linked to each child based on district (or cluster, where feasible) of residence.

3.2 Data sources

Child and household data: Multiple Indicator Cluster Survey (MICS) 2019 (or 2025 if available). The survey provides anthropometric measurements, child feeding information, and household socioeconomic characteristics.

Agricultural production data: Bangladesh Bureau of Statistics (BBS) Yearbook of Agricultural Statistics. Annual district-level yield data for rice, wheat, maize, and potato over approximately 20 years will be used to construct production indicators and long-term anomalies.

Climate data: Bangladesh Meteorological Department (BMD) climate datasets providing temperature and rainfall measures. Climate indicators will be aggregated to district level and aligned temporally with each child’s 1,000-day exposure window.

3.3 Exposure: Climate shock

Climate shock will be operationalized using agro-climatic indicators during the 1,000-day window covering in utero exposure and early childhood.

Growing degree days (GDD)

$$GDD = \sum_{t=1}^T \max \left(\frac{T_{max,t} + T_{min,t}}{2} - T_{base}, 0 \right)$$

GDD captures accumulated heat exposure relevant for crop growth conditions (McMaster and Wilhelm, 1997).

Heating degree days (HDD)

$$HDD = \sum_{t=1}^T \max(T_{mean,t} - T_{threshold}, 0)$$

HDD measures excessive heat exposure above biologically or agronomically relevant thresholds ([McMaster and Wilhelm, 1997](#)).

Extreme rainfall and rainfall deficit

Rainfall shocks will be measured using deviations from long-term district means:

$$Rainfall\ Anomaly_{dt} = \frac{R_{dt} - \bar{R}_d}{\bar{R}_d}$$

where R_{dt} denotes rainfall in district d during year t , and $\bar{R}_d$ is the historical district mean. Indicators for extreme rainfall and rainfall deficit will be constructed using percentile-based thresholds or standardized anomalies to capture both excessive precipitation and drought risk.

Season-specific exposures during pregnancy and early childhood will be assigned by linking spatially referenced survey clusters to district-level climate data.

3.4 Agricultural production

Agricultural production will be measured as total annual yield of four major crops: rice, wheat, maize, and potato. For each district and year, aggregate crop production will be calculated. To capture deviations from normal production levels, district-specific anomalies will be computed:

$$Production\ Anomaly_{dt} = \frac{Y_{dt} - \bar{Y}_d}{\bar{Y}_d}$$

where Y_{dt} represents total crop yield in district d during year t , and $\bar{Y}_d$ is the long-term district average over the 20-year reference period. Production measures occurring within each child's 1,000-day window will be aggregated to reflect cumulative agricultural exposure.

3.5 Dietary diversity

Dietary diversity will be assessed using the Minimum Dietary Diversity (MDD) indicator, following the guidelines established by the World Health Organization (WHO) and the United Nations Children's Fund (UNICEF) ([World Health Organization and United Nations Children's Fund \(UNICEF\), 2021](#)). Reported food and liquid consumption from the day and night preceding the survey will be aggregated into eight standard food groups: (1) breastmilk; (2) grains, roots, and tubers; (3) legumes and nuts; (4) dairy products; (5) flesh foods; (6) eggs; (7) vitamin A-rich

fruits and vegetables; and (8) other fruits and vegetables. A dietary diversity score (ranging from 0 to 8) will be computed for each child. Consequently, MDD will be constructed as a binary variable, taking the value of 1 if the child consumed items from at least five of the eight food groups, and 0 otherwise. This measure reflects diet quality and micronutrient adequacy, serving as a critical pathway linking agricultural production to child nutritional outcomes.

3.6 Outcome: Child nutritional status

Child nutritional status will be assessed using standardized anthropometric indicators based on the World Health Organization (WHO) child growth standards ([World Health Organization, 2006](#)). These measures compare a child’s height and weight with those of a healthy reference population of the same age and sex.

The analysis will focus on three commonly used indicators:

- **Stunting (Height for Age)** — reflects chronic undernutrition and long term growth faltering.
- **Wasting (Weight for Height)** — reflects acute undernutrition and recent weight loss.
- **Underweight (Weight for Age)** — a composite indicator capturing elements of both chronic and acute undernutrition.

Following standard WHO definitions, children whose anthropometric measurements fall below -2 standard deviations from the WHO reference median will be classified as stunted, wasted, or underweight, respectively ([World Health Organization, 2006](#)). These binary indicators will serve as the primary outcome variables in the analysis.

3.7 Covariates

The following child and household-level covariates will be included to control for socioeconomic and demographic factors:

- Household wealth index
- Maternal education
- Child age
- Child sex
- Place of residence

- Maternal age
- Household size etc.

3.8 Conceptual framework

The analytical framework is guided by the following hypothesized relationships:

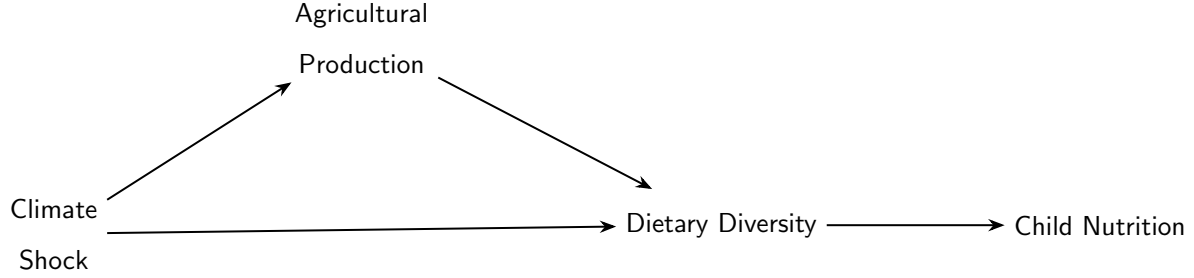


Figure 1: Impact of Climate Shock on Child Nutrition

3.9 Modeling approach

Given the multilevel structure of the data and the sequential nature of exposures, mediators, and outcomes, we will employ a multilevel generalized structural equation modeling (GSEM) framework to estimate spatially structured associative pathways. This approach allows decomposition of total associations into pathway-specific indirect effects while accounting for clustering at the district level and spatial dependence.

Notation

Let:

- C_d = climate shock exposure in district d during the 1,000-day window
- A_d = agricultural production anomaly in district d
- D_{id} = Minimum dietary diversity of child i in district d
- Y_{id} = binary nutritional outcome (stunting, wasting, underweight) for child i in district d
- $\mathbf{X}_{id}$ = vector of child- and household-level covariates

Structural equations

1. Agricultural production equation:

$$A_d = \alpha_0 + \alpha_1 C_d + u_d$$

where u_d captures district-level random effects.

2. Dietary diversity equation:

$$\text{logit}(D_{id}) = \beta_0 + \beta_1 C_d + \beta_2 A_d + \mathbf{X}_{id}\boldsymbol{\gamma} + v_d + \epsilon_{id}$$

where v_d is a district-level random effect and ϵ_{id} is the child-level residual.

3. Nutritional outcome equation:

$$\text{logit}(P(Y_{id} = 1)) = \theta_0 + \theta_1 C_d + \theta_2 A_d + \theta_3 D_{id} + \mathbf{X}_{id}\boldsymbol{\delta} + w_d$$

with w_d representing district-level spatially structured random effects.

Spatial dependence

District-level random effects w_d in the outcome equation will be modeled using a conditional autoregressive (CAR) structure:

$$w_d \sim CAR(W, \tau)$$

where W is the adjacency matrix of districts and τ is the precision parameter ([Besag, 1974](#)). This allows capturing spatial autocorrelation in both exposures and outcomes.

4 Time frame

The proposed timeline for completion of this research is as follows:

Table 1: Proposed time schedule for the thesis

Month	Task
June, 2026	Data preparation and variable construction with exposure window
July, 2026	Exploratory analysis and development of methodological framework
August, 2026	Model specification and estimation with spatial effects
September, 2026	Validation and diagnostic checking
October, 2026	Interpretation of findings and report writing